

JEE Main - 3 | JEE - 2023 (Gen 1 & 2)

Date: 10/08/2021

Maximum Marks: 300

Timing: 04:00 PM - 09:00 PM

Duration: 3.0 Hrs

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **300**.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is an **integer** ranging from 0 to 9. (both inclusive)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) :

Roll Number :

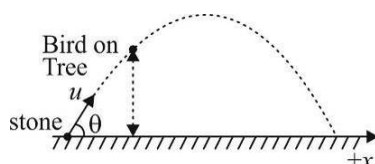
OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

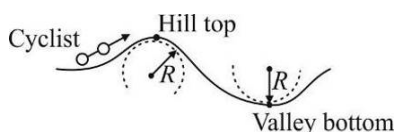
PART - I : PHYSICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

1. A stone is thrown with speed u at an angle $\theta = 37^\circ$, in such a manner that it can hit a bird sitting at the top of a tree. As the stone was about to hit the bird, the bird starts flying away horizontally along positive x direction with speed 4 m/s. But even then, after sometime the stone hits the bird. Find the value of u .

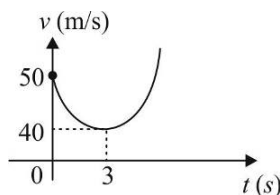


- (A) 4 m/s (B) 5 m/s (C) 3 m/s (D) 3.2 m/s
2. A cyclist travels over a circular hill and a circular valley with the same radius, as shown in the figure. The speed of cyclist doubles as he goes from top of the hill to the bottom of valley. Mass of the cyclist (along with the cycle) is 60 kg. The normal force on cycle from ground is zero at top of the hill. Find the normal force's value as the cycle passes through the bottom of valley. (Consider $g = 10 \text{ m/s}^2$)

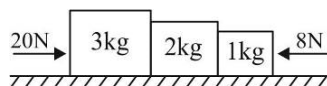


- (A) 3000 N (B) 1800 N (C) 2400 N (D) 3600 N
3. A particle initially at rest starts moving in a circle of radius 2m. It's angular acceleration varies with time as $\alpha = 4 - 2t$, where α and t are in SI units. Find the speed of the particle, when the total acceleration of particle becomes perpendicular with the velocity vector.
- (A) 4 m/s (B) 16 m/s (C) 8 m/s (D) 32 m/s
4. Ball A and Ball B are thrown from ground at angles 37° and 45° respectively from horizontal. If both balls have same range on the ground, then the ratio of initial velocities of Ball A to Ball B is:
- (A) $\frac{24\sqrt{6}}{5}$ (B) $\frac{5\sqrt{6}}{12}$ (C) $\frac{12}{5\sqrt{6}}$ (D) $\frac{5\sqrt{6}}{24}$
5. A boy throws a ball vertically upwards with velocity 30 m/s. At the instant of throw, wind starts blowing along horizontal direction, imparting a constant acceleration of 2 m/s^2 to the ball along horizontal direction. Assuming no effect of wind in the vertical direction, how fast the boy should run so as to catch the ball? Consider boy runs with a constant velocity. (Consider $g = 10 \text{ m/s}^2$)
- (A) 4 m/s (B) 3 m/s (C) 12 m/s (D) 6 m/s

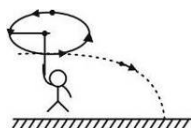
6. A ball is projected at certain angle θ from the top a building of height 80 m. The speed of the ball as a function of time is as shown. Find the angle from horizontal, at which the ball strikes the ground. (Consider $g = 10 \text{ m/s}^2$)



- (A) $\tan^{-1} \frac{5}{4}$ (B) $\tan^{-1} \frac{3}{4}$ (C) $\tan^{-1} \frac{4}{5}$ (D) $\tan^{-1} \frac{4}{3}$
7. A hunter sitting on ground aims at a monkey hanging on a tree. The height of the monkey from ground is 10 m. The shortest distance between monkey and hunter is 400 m. As soon as the hunter fires a bullet, the monkey drops from the branch. If the speed of the bullet is 500 m/s, find the height above the ground where the bullet will hit the monkey? (Consider $g = 10 \text{ m/s}^2$)
- (A) 4 m (B) 3.2 m (C) 6.8 m (D) 6 m
8. A girl can throw a stone to a maximum horizontal distance of 40 m on earth. To what maximum height can she throw the same stone on moon where free fall acceleration is one-sixth that on the earth ?
- (A) 40 m (B) 120 m (C) 60 m (D) 240 m
9. Three blocks of mass 1 kg, 2 kg and 3 kg are lying in contact on a smooth horizontal surface. Two horizontal forces of magnitudes 20 N and 8 N are applied as shown. Then the contact force between the 3 kg and 2 kg block has magnitude:



- (A) 4 N (B) 6 N (C) 10 N (D) 14 N
10. A particle has initial velocity $4\hat{i} + 3\hat{j}$ and a constant acceleration $-12\sqrt{3}\hat{i} + b\hat{j}$. What should be the value of 'b' so that particle is able to pass again through its original position?
- (A) $9\sqrt{3}$ (B) $-16\sqrt{3}$ (C) $16\sqrt{3}$ (D) $-9\sqrt{3}$
11. A ball is projected with initial speed 10 m/s at an angle 37° from horizontal. Then the radius of curvature of ball's path as it passes through the top most point is: (Consider $g = 10 \text{ m/s}^2$)
- (A) 6.4 m (B) 9.6 m (C) 4.8 m (D) 1.8 m
12. A boy rotates a stone using a string in a horizontal circle of radius 0.4 m. Mass of stone is 0.5 kg and it rotates with constant angular velocity. At some point the boy releases the string and the stone falls following a projectile path. Horizontal range of the stone is 10 m and time taken during the fall is 0.5 seconds. Find tension in the string while the stone was being rotated. (Consider $g = 10 \text{ m/s}^2$)



- (A) 1000 N (B) 250 N (C) 500 N (D) 125 N

13. A fighter plane dives in such a way that its horizontal velocity remains constant and acceleration in vertical direction becomes 'g' downwards. During the dive, the weight of a mass m measured by a weighing machine inside the plane is:

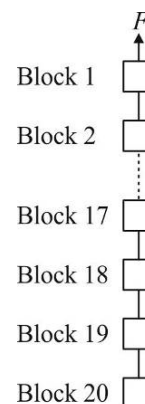
(A) $2mg$ (B) mg (C) $0.5mg$ (D) 0

14. Block A of mass 3 kg is placed over Block B of mass 5 kg. Block B rests on a rough horizontal surface. Coefficient of friction between block A and block B is 0.2 and between block B and ground is 0.6. Then the maximum value of force that can be applied on block B along horizontal direction, so that both blocks move together is: (Consider $g = 10 \text{ m/s}^2$)

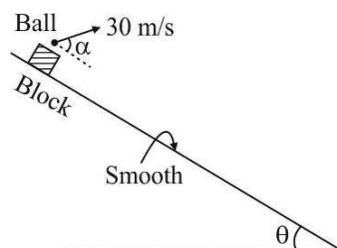
(A) 48 N (B) 64 N (C) 54 N (D) 70 N

15. 20 blocks each of mass $m = 5 \text{ kg}$ are connected using massless strings and are hanging in vertical direction. An external force F of magnitude 1500 N is applied in upwards direction over the first block, as shown in the diagram. Find the tension in the rope between 17th and 18th block. (Consider $g = 10 \text{ m/s}^2$)

(A) 225 N
(B) 300 N
(C) 375 N
(D) 150 N

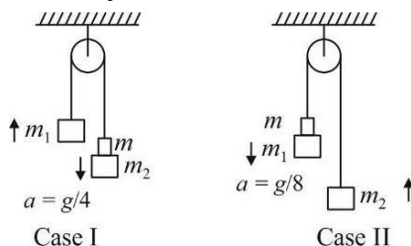


16. From an inclined plane, a ball is projected with speed 30 m/s making an angle $\alpha = 53^\circ$ with incline. At the same instant a block also starts sliding from the same initial position, as shown in the diagram. The incline is smooth and makes an angle $\theta = 37^\circ$ from the horizontal. The separation between block and the ball, when the ball hits the incline for the first time will be: (Consider $g = 10 \text{ m/s}^2$)



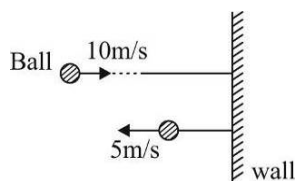
(A) 48 m (B) 108 m (C) 36 m (D) 144 m

17. Two masses m_1 and m_2 tied to a string are hanging over a light frictionless pulley. If a block of mass $m = 3 \text{ kg}$ is placed over m_2 , the acceleration of m_2 becomes $g/4$ downwards. Instead if the same block m is placed over m_1 , acceleration of m_1 becomes $g/8$ downwards. Find the ratio m_2 / m_1 .

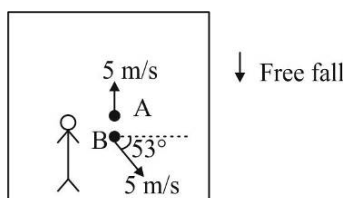


(A) $\frac{6}{7}$ (B) $\frac{5}{6}$ (C) $\frac{7}{6}$ (D) $\frac{6}{5}$

18. A ball of mass $m = 5\text{kg}$, moving with velocity 10 m/s hits a wall and returns back. The return speed is 5 m/s and the ball remains in contact with the wall for 0.1 s . The average force applied by the wall on the ball is:



- (A) 1000 N (B) 500 N (C) 250 N (D) 750 N
19. A car enters a plane horizontal circular road of radius 100 m at a speed of 20 m/s . The car also has a tangential acceleration a_t . Mass of car is 600 kg and the coefficient of friction between car and road is 0.8 . Find the maximum possible value of a_t so that the car does not slip the moment it enters the road. (Consider $g = 10\text{ m/s}^2$)
- (A) $4\sqrt{3}\text{ m/s}^2$ (B) 8 m/s^2 (C) $4\sqrt{5}\text{ m/s}^2$ (D) 6 m/s^2
20. In a freely falling elevator a boy throws ball A upwards with speed 5 m/s . At the same instant he also throws ball B at an angle 53° below horizontal with speed 5 m/s . The speeds and angle of projection are with respect to the elevator. The ball A hits the ceiling in time $t = 0.2\text{ sec}$ and in same time ball B also hits the floor (and not side walls). Assuming that boy throws both the balls from the level of his head, find the ratio of boy's height to elevator's height.



- (A) $\frac{1}{3}$ (B) $\frac{4}{9}$ (C) $\frac{5}{9}$ (D) $\frac{2}{3}$

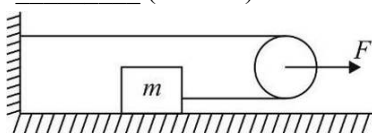
SPACE FOR ROUGH WORK

SECTION-2

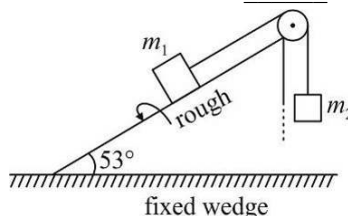
This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 9 (both inclusive).

21. In a game of archery, an archer and target are $80\sqrt{3}$ m apart and are on same horizontal level. The archer shoots arrows with speed 40 m/s to hit the target. Some of the arrows which hit the target take time t_1 while some take time t_2 , with $t_1 > t_2$. Find $\left(\frac{t_1}{t_2}\right)^2$.

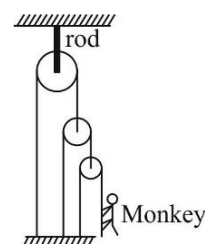
22. In the figure shown, if the pulley moves with an acceleration of 2 m/s^2 upon application of force F , then acceleration of the block will be _____ (in m/s^2).



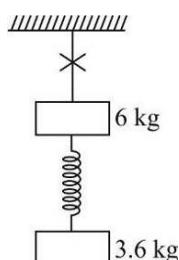
23. In the arrangement shown in the figure, coefficient of friction between m_1 and wedge is 0.6, angle of wedge with horizontal is 53° , and mass of m_2 is 2.2 kg. The minimum value of m_1 for which it starts sliding down the wedge is x kg. Then the value of x is _____.



24. In the arrangement shown all string are massless and have same breaking strength of 300 N. The monkey has mass of 5 kg. Assuming that rod holding the pulley doesn't break, then the maximum acceleration with which monkey can climb up without breaking any string is $n \text{ m/s}^2$. The value of n is _____. (Consider $g = 10 \text{ m/s}^2$)



25. System shown in the figure is in equilibrium and at rest. The magnitude of acceleration of block of mass 6 kg as soon as the upper string is cut is $x^2 \text{ m/s}^2$. The value of x is _____. (Consider $g = 10 \text{ m/s}^2$)



SPACE FOR ROUGH WORK

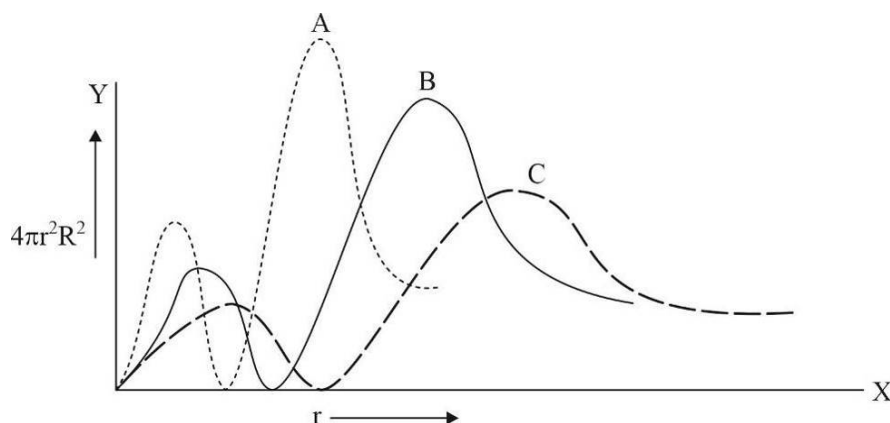
PART - II : CHEMISTRY**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- In which of the following sets central atom of each member involve sp^3 hybridisation :
 (A) IO_4^- , ICl_4^- , IF_4^+ (B) XeO_3 , XeO_4 , XeF_4
 (C) SO_3^{2-} , SO_4^{2-} , NH_4^+ (D) PCl_4^+ , BF_4^- , ICl_4^-
- In which of the following groups all the members have linear shape :
 (A) NO_2^+ , N_3^- , $SnCl_2$ (B) N_3^- , I_3^- , C_2H_2
 (C) XeF_2 , C_2H_2 , SO_2 (D) CO_2 , $BeCl_2$, $SnCl_2$
- In O_2 molecules which of the following molecular orbital are lower in energy than $\sigma 2p_z$ orbital :
 I. π_{2px} II. $\sigma 2s$ III. σ_{2s}^* IV. π_{2py}
 (A) I, II and III (B) II and III (C) II, III and IV (D) All of the above
- Which of the following statement is incorrect?
 (A) Among O_2^+ , O_2 and O_2^- , the stability decreases as $O_2^+ > O_2 > O_2^-$
 (B) He_2 molecule does not exist as the bonding and anti-bonding electrons cancel each other
 (C) C_2 , C_2^{2-} and Li_2 are diamagnetic
 (D) In F_2 molecule, the energy of σ_{2p_z} is more than π_{2p_x} and π_{2p_y}
- The compound which contains both ionic and covalent bonds is :
 (A) CH_4 (B) H_2 (C) KCN (D) KCl
- If a molecule MX_3 has zero dipole moment, the sigma bonding orbitals used by M (atomic number < 21) are :
 (A) pure p (B) sp hybridised (C) sp^2 hybridised (D) sp^3 hybridized
- Which of the following order is correct for the property mentioned in brackets?
 (A) $S^{2-} > Cl^- > K^+ > Ca^{2+}$ (Ionisation energy)
 (B) $C < N < F < O$ (2^{nd} Ionisation energy)
 (C) $B > Al > Ga > In < Tl$ (Atomic size)
 (D) $Na^+ > Li^+ > Mg^{2+} > Be^{2+} > Al^{3+}$ (Ionic radius)
- Which of the following is the smallest in size?
 (A) N^{3-} (B) O^{2-} (C) F^- (D) Na^+
- Which of the following orders is incorrect with respect to the property indicated?
 (A) $NH_3 > PH_3 > AsH_3$ (bond angle)
 (B) $Li < Be < B < C$ (first ionisation energy)
 (C) $Al_2O_3 < MgO < Na_2O < K_2O$ (basic character)
 (D) $Li^+ < Na^+ < K^+ < Cs^+$ (ionic radius)

10. The atomic number of elements A, B, C and D are $Z-1$, Z , $Z+1$, $Z+2$ respectively. If B is a noble gas, choose the correct option:
 (i) (A) has higher electron affinity
 (ii) (C) exist in +2 electrovalency
 (iii) (D) is an alkaline earth metal
 (A) (i) and (ii) (B) (ii) and (iii) (C) (i) and (iii) (D) (i), (ii) and (iii)
11. If the dipole moment of AB molecule is given by $1.2D$ and A-B bond length is $1\text{\AA}$. Then % ionic character of the bond is : [Given : $1\text{ debye} = 10^{-18}\text{esu.cm}$]
 (A) 75 (B) 50 (C) 60 (D) 25
12. Which of the following molecules has maximum number of 90° bond angles?
 (A) PCl_5 (B) SF_6 (C) BrF_5 (D) XeF_4
13. Spin only magnetic moment 2.84 BM is given by :
 (Atomic Number of Ni = 28, Ti = 22, Cr = 24, Co = 27)
 (A) Ni^{2+} (B) Ti^{3+} (C) Cr^{2+} (D) Co^{2+}
14. An electron in an atom jumps in such a way that its kinetic energy changes from x to $\frac{x}{9}$. The change in its potential energy (magnitude) will be :
 (A) $\frac{x}{9}$ (B) $\frac{16x}{9}$ (C) $9x$ (D) $\frac{9x}{16}$
15. The angular momentum of an electron in He^+ moving in an orbit is h/π . The de-Broglie wavelength associated with electron is : [a_0 is radius of first Bohr's orbit of H-atom]
 (A) $2\pi a_0$ (B) πa_0 (C) $4\pi a_0$ (D) $\frac{\pi a_0}{2}$
16. In case of photoelectric effect, slope of V_0 vs ν curve is :
 (Where V_0 - stopping potential, ν = subjected frequency)
 (A) e (B) $\frac{h}{e}$ (C) ϕ (D) h
17. The maximum probability of finding electron in the d_{xy} orbital is :
 (A) along the x-axis
 (B) along the y-axis
 (C) at the angle of 45° from the x-axis and y-axis
 (D) at the angle of 90° from the x-axis and y-axis
18. Bond dissociation energy of Cl_2 is 240 kJ/mol . The longest wavelength of photon that can break this bond would be : [$N_A = 6 \times 10^{23}$, $h = 6.6 \times 10^{-34}\text{ J/s}$]
 (A) $4.95 \times 10^{-7}\text{ m}$ (B) $9.9 \times 10^{-7}\text{ m}$ (C) $4.95 \times 10^{-6}\text{ m}$ (D) $9.9 \times 10^{-6}\text{ m}$

19. Out of the following, which is the correct match for the radial probability of finding the electron for 2s orbital:



- (A) A – H, B – He^+ , C – Li^{2+} (B) A – He^+ , B – H, C – Li^{2+}
(C) Can't say (D) A – Li^{2+} , B – He^+ , C – H
20. Among the following the molecule with the highest dipole moment is :
- (A) CH_3Cl (B) CH_2Cl_2 (C) CHCl_3 (D) CCl_4

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 9 (both inclusive).

21. If value of spin quantum number(s) = $-1/2$, 0, $+1/2$ then find out number of elements in first period in the new form of periodic table if all other rules of electronic configurations remain same.
22. First and second ionization energies of magnesium are 7.646 and 15.035 eV respectively. The amount of energy in kJ/mol needed to convert all the atoms of Magnesium into Mg^{2+} ions present in 12 mg of magnesium vapour is : [Given : $1\text{ eV} = 96.5\text{ kJ mol}^{-1}$, Atomic mass of Mg = 24]
23. Number of electrons having $\ell + m = 0$ in Mg ($z = 12$) is :
24. The sum of number of sigma and pi bonds formed between two carbon atoms in CaC_2 are :
25. The number of paramagnetic species among the following is :
 O_2 , CO , N_2 , C_2 , CsO_2 , BaO_2 , CO_2

SPACE FOR ROUGH WORK

PART - III : MATHEMATICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- The expression, $\frac{\tan\left(\frac{3\pi}{2} - \alpha\right) \cos\left(\frac{3\pi}{2} - \alpha\right)}{\cos(2\pi - \alpha)} + \cos\left(\alpha - \frac{\pi}{2}\right) \sin(\pi - \alpha) + \cos(\pi + \alpha) \sin\left(\alpha - \frac{\pi}{2}\right)$, $\alpha \neq \frac{n\pi}{2}, n \in I$ when simplified reduces to:
 (A) zero (B) 1 (C) -1 (D) None of these
- The real number x and y satisfy the equation $xy = \sin(2t)$ and $\frac{x}{y} = \tan(t)$ where $0 < t < \frac{\pi}{2}$. The value of $x^2 + y^2$, is:
 (A) $\sqrt{2}$ (B) 1 (C) 2 (D) 4
- Let z_0 be a complex root of the quadratic equation $x^2 + x + 1 = 0$. If $z = 3\sqrt{3} + 11iz_0^{363} - 8iz_0^{93}$, then $|z|$ equals :
 (A) $3\sqrt{3}$ (B) 6 (C) 0 (D) None of these
- Let $A = \left\{ \theta \in (-\pi, \pi) : \frac{5 + 4i \sin \theta}{3 - 2i \sin \theta} \text{ is purely imaginary} \right\}$. The number of elements in A is :
 (A) 0 (B) 2 (C) 4 (D) 6
- The expression $f(x) = \frac{2 \sin^2 x - 1}{\cos x} + \frac{\cos x(1 + 2 \sin x)}{1 + \sin x}$ wherever defined simplifies to:
 (A) $\tan^2 x$ (B) $\tan x$ (C) $1 - \cos x$ (D) $1 + \sin^2 x$
- Exact value of $\frac{\cos 21^\circ - \sin 21^\circ}{\sin^2 57^\circ - \sin^2 33^\circ}$ is equal to:
 (A) 2 (B) $-\sqrt{2}$ (C) 1 (D) $\sqrt{2}$
- Let $z = \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^{15} - \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)^{15}$. If $R(z)$ and $I(z)$ respectively denote the real and imaginary parts of z , then :
 (A) $R(z) > 0$ (B) $I(z) < 0$
 (C) $R(z) - I(z) = 0$ (D) $R(z) + I(z) > 0$
- Let $(1 + \sqrt{3}i)^{12} = 64(x + iy) \dots (i = \sqrt{-1})$, where x and y are real numbers, then $x - y$ equals :
 (A) 0 (B) 64 (C) 128 (D) 2^{12}

9. If $A = 10^\circ$, then the value of expression $\left(\frac{\sec 2A}{(1 - 4\sin^2 2A)} + \frac{\operatorname{cosec} A}{(4\cos^2 A - 1)} \right)^2$ is:
 (A) 6 (B) 8 (C) 16 (D) 24
10. Exact value of $\sec 10^\circ - \tan 10^\circ - \tan 40^\circ$ is equal to:
 (A) $\sqrt{3}$ (B) 2 (C) 0 (D) $\frac{-1}{\sqrt{3}}$
11. Let $\alpha, \beta \in \mathbb{C}$ and roots of the equation $\alpha z^2 + z + \beta = 0$ lie on $|z| = \sqrt{2}$ where z is a complex number then the value of $|\beta|^2$ in terms of complex number α is :
 (A) $2|\alpha|^2$ (B) $3|\alpha|^2$ (C) $4|\alpha|^2$ (D) $6|\alpha|^2$
12. Let α and β be two roots of equation $x^2 - x + 1 = 0$ then $\alpha^{30} - \beta^{45}$ equal to :
 (A) 0 (B) 1 (C) -1 (D) None of these
13. Let z_1 and z_2 be two complex numbers satisfying $|z_1| = 9$ and $|z_2 - 12 - 5i| = 4$ then maximum value of $|z_1 - z_2|$ is :
 (A) 13 (B) 20 (C) 26 (D) None of these
14. If $\frac{\bar{z} - \alpha}{\bar{z} + \alpha}$ ($\alpha \in \mathbb{R}$) is a purely imaginary number and $|z| = 2$, then value of α is :
 (A) $\sqrt{2}$ (B) 2 (C) $\frac{1}{2}$ (D) 1
15. The maximum value of $(7\cos\theta + 24\sin\theta) \times (7\sin\theta - 24\cos\theta)$ for every $\theta \in \mathbb{R}$.
 (A) 25 (B) 625 (C) $\frac{625}{2}$ (D) $\frac{625}{4}$
16. The sum of distinct solution of the equation $4\sin\theta\cos\theta - 2\cos\theta - 2\sqrt{3}\sin\theta + \sqrt{3} = 0$ in $[0, 2\pi]$ is:
 (A) 2π (B) π (C) $\frac{13\pi}{6}$ (D) $\frac{17\pi}{6}$
17. The number of distinct solution of $\sin 5\theta \cdot \cos 3\theta = \sin 9\theta \cdot \cos 7\theta$ in $\left[0, \frac{\pi}{4}\right]$ is:
 (A) 4 (B) 5 (C) 8 (D) 9
18. If A and B are complimentary angles, then:
 (A) $\left(1 + \tan \frac{A}{2}\right)\left(1 + \tan \frac{B}{2}\right) = 2$ (B) $\left(1 + \cot \frac{A}{2}\right)\left(1 + \cot \frac{B}{2}\right) = 2$
 (C) $\left(1 + \sec \frac{A}{2}\right)\left(1 + \operatorname{cosec} \frac{B}{2}\right) = 2$ (D) $\left(1 - \tan \frac{A}{2}\right)\left(1 - \tan \frac{B}{2}\right) = 2$
19. Let z be a complex number such that $|z| - z = 5 + \sqrt{5}i$ (where $i = \sqrt{-1}$) then $|z|$ is equal to :
 (A) $\sqrt{2}$ (B) 2 (C) $2\sqrt{2}$ (D) 3

20. If $z = \frac{1}{2}(i\sqrt{3} - 1)$, then the value of $(z - z^2 + 2)(2 - z + z^2)$ is:
 (A) 3 (B) 7 (C) -1 (D) 5

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 9 (both inclusive).

21. If the expression, $2\cos 10^\circ + \sin 100^\circ + \sin 1000^\circ + \sin 10000^\circ$ is simplified, then it simplifies to $a \cos b^\circ$ where a is natural number and b is acute angle, then $b - a$ is _____.
22. Let $f(x) = 11 - 8\sin x - 2\cos^2 x$. If the maximum and minimum values of $f(x)$ are denoted by M and m respectively, then $\frac{M+8}{m}$ has the value equal to:
23. If m and n are positive integers satisfying $1 + \cos 2\theta + \cos 4\theta + \cos 6\theta + \cos 8\theta + \cos 10\theta = \frac{\cos m\theta \cdot \sin n\theta}{\sin \theta}$, then $(n - m)$ is equal to _____.
24. If α is an imaginary fifth root of unity, then $4\log_4 \left| 1 + \alpha + \alpha^2 + \alpha^3 - \frac{1}{\alpha} \right|$ is _____.
25. Let $w = \frac{\sqrt{3} + i}{2}$ and $P = \{w^n : n = 1, 2, 3, \dots\}$. Further $H_1 = \left\{ z \in C : \operatorname{Re} z > \frac{1}{2} \right\}$ and $H_2 = \left\{ z \in C : \operatorname{Re} z < \frac{-1}{2} \right\}$, Where C is the set of all complex numbers. If $z_1 \in P \cap H_1$, $z_2 \in P \cap H_2$ and O represents the origin, then least positive value of $\angle z_1 O z_2$ is $\frac{a\pi}{b}$ (where a and b coprime natural numbers), then $a + b$ is _____.

SPACE FOR ROUGH WORK

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